Control Stack

Control system hardware for high-powered rocket targeting 30,000ft



For more of my projects, visit: gnd12.github.io

General Overview

The control stack consists of 2 modules:

- Canards (top) to control rocket roll rate.
- Airbrakes (bottom) to control drag and control rocket apogee

Both modules are 6.125” outer diameter, and must interface with composite body tubes on top and bottom

Each module can be integrated and serviced separately.





Preliminary design decisions

My design is the second iteration of the control stack.

Design targets:

1. Reduce overall weight & length of module
2. Increase control authority
3. Streamline manufacturing and integration efforts

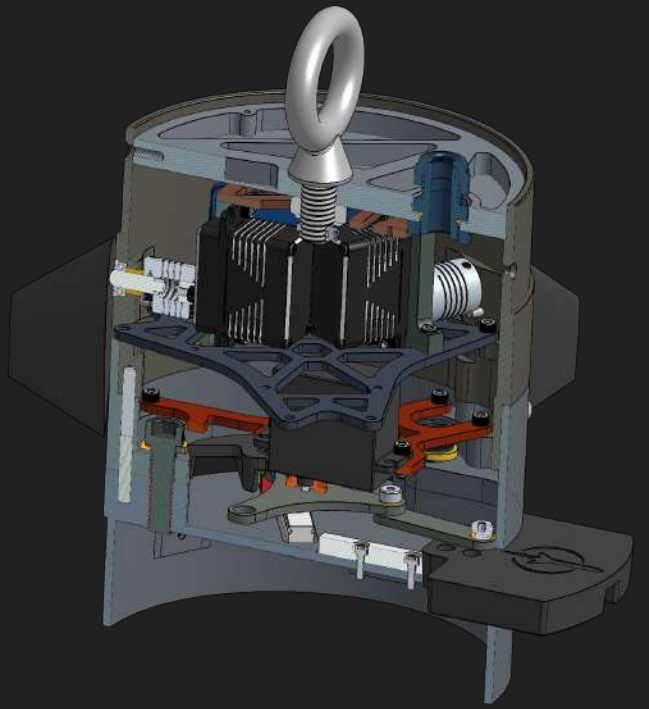
← Comparison between V1 (10.1 lb, 13.5” long, 93 parts) and my V2 (6.2 lb, 9.5” long, 76 parts)

1. Weight & Length reduction

Length reduction primarily achieved by tight packaging: overall module length constrained by length of servos only

Weight reduction by aggressive contouring & pocketing of servo plates.

Nominal housing wall thickness is 0.125”. Design was iterated over multiple FEA cycles to remove as much material as possible. Alignment between modules are achieved with dowel pins.

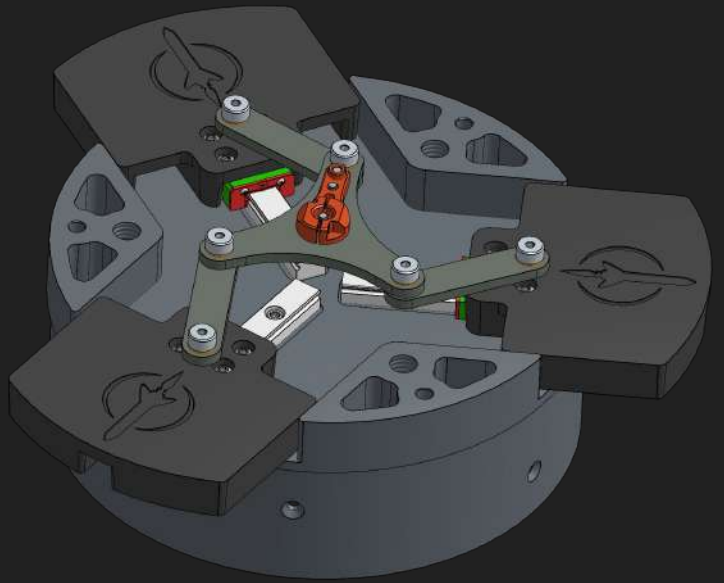


1. Weight & Length reduction

Airbrake and canard blades are forged carbon fiber, 30% lighter than aluminum.

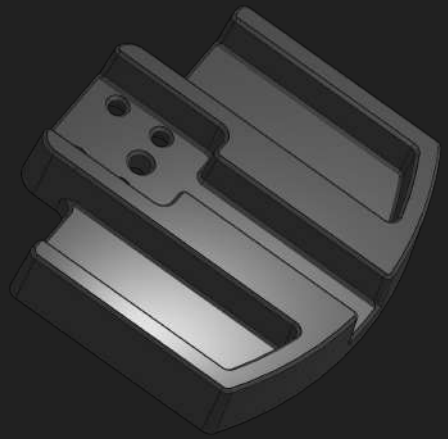
Airbrake uses tri-bar linkage for even actuation from a single central servo. Full blade deployment achieved with 75 degrees servo actuation, allowing for fine control.

Dimensions of each linkage are optimized: small enough to be light, but constrained by the smallest size of bushing and shoulder screw available from McMaster-Carr





Initial blade design with aggressive weight reduction pocketing on the bottom



Revised blade design for ease of mold removal

Blade, forged & clear coated →





Blade_width_calculator.ipynb

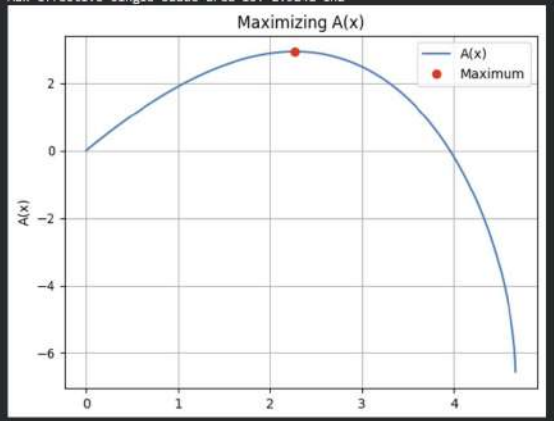
File Edit View Insert Runtime Tools Help

Commands + Code + Text ▷ Run all

```
x_max = x_vals[max_idx]
A_max = A_vals[max_idx]

import datetime
print(datetime.datetime.now())
print(f"Blade width is {x_max:.4f} in")
print(f"Linear ROM of blade is {A_max/x_max:.4f} in")
print(f"Max effective single blade area is: {A_max:.4f} in2")
# Optional: plot
plt.plot(x_vals, A_vals, label='A(x)')
plt.plot(x_max, A_max, 'ro', label='Maximum')
plt.xlabel('x')
plt.ylabel('A(x)')
plt.title('Maximizing A(x)')
plt.legend()
plt.grid(True)
plt.show()
```

```
2026-07-22 22:22:20.044685
Blade width is 2.2697 in
Linear ROM of blade is 1.2883 in
Max effective single blade area is: 2.9241 in2
```



I wrote a little script to find the optimal size of airbrake blades

2. Control authority increase

For airbrakes, this is an easy problem:

- Given a rocket diameter, there exists one blade width and length that will give the maximum area → maximum drag

For canards, however, this is a harder problem:

- Cannot simply increase blade size (rocket stability margin decreased)
- Can increase actuation angle, but must not cause vortex shedding (affecting tail-fin flow)

Welcome to: canard blade design 101, and why rocket science is all about trade-offs

A good design must have

- A small cross-sectional airfoil area to minimize supersonic wave drag
- A strong mechanical connection to pivot shaft
- A center of pressure aft of pivot point to allow aerodynamic feathering



Design 1: NACA 0006, most streamlined, but too thin for a secure mechanical connection

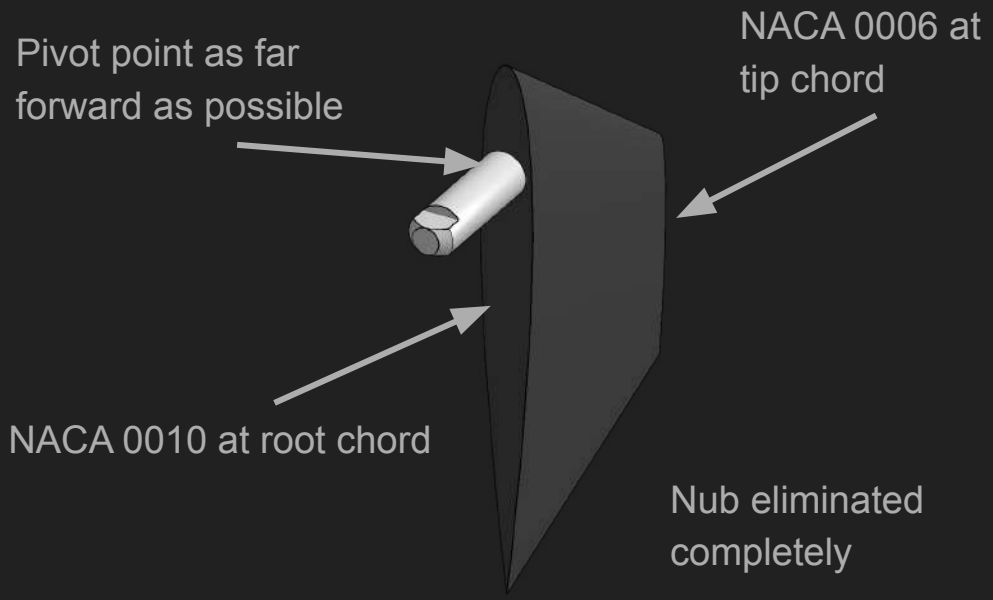


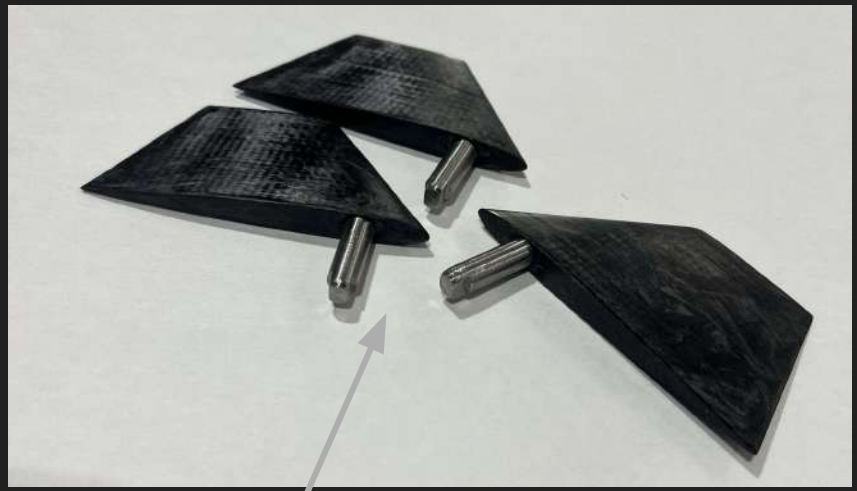
Design 2: NACA 0010, very strong mechanical connection, but 60% thicker than design 1 and 2.5x more drag



Design 3: NACA 0006, and strengthen the connection point with a little nub. BUT the nub pushes the center of pressure in front of pivot point!

Middle ground design





Pioneering technique: forged CF overmolded directly on shaft - reduce part count & manufacturing complexity

3. Streamline manufacturing & integration

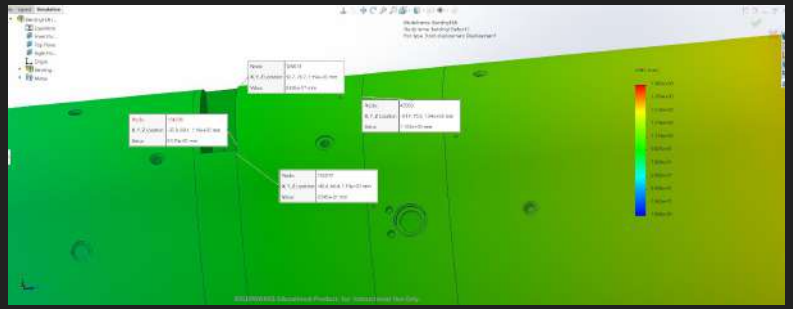
All servo plates & linkages designed for water jetting

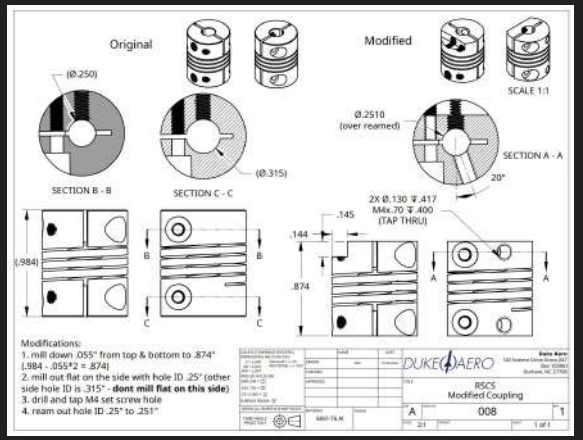
Canard-servo coupling modified from cheap off-the-shelf component

New radial bolt interface between 2 modules - allow for external integration

Also shortest bolted lap joint on the rocket - demanded lots of analysis to verify strength







Manufacturing saga

The hard lesson learned: thin-walled housing is light and strong, but difficult to manufacture



Attempt 1: Crashed drill. Always verify tool holder in CAM is the same length as in real life



Attempt 2: Tried light clamping to preserve surface finish, but got too cocky with speeds and part flew out of vise



Attempt 3: Used lathe chuck for more points of contact. Thin flexible wall acted like a spring and part flew out of chuck



Attempt 4: Workholding with fixture plate. Success!

More manufacturing & integration photos



